

ID:

Name:

**Brac University**

Semester: Spring 2023

Course Code: CSE250

Circuits And Electronics

Section: 05 & 06

Faculty: SHS

Set

B

Assessment: Quiz 3

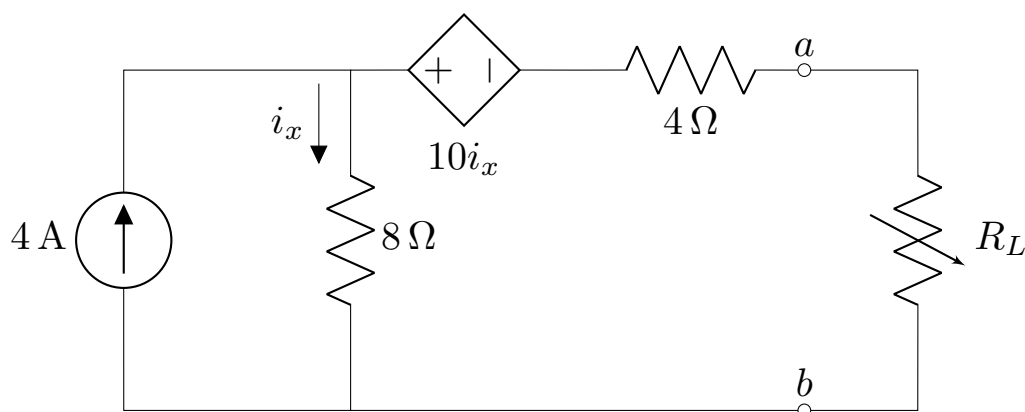
Duration: 30 minutes

Date: April 18, 2023

Full Marks (incl. bonus 4): 20

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to invigilator.
- ✓ All **1 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Bonus questions are indicated as “(bonus)” along with allotted marks.
- ✓ Write your answers inside the indicated boxes (where applicable). If you run out of room, continue on the back page.
- ✓ Symbols have their usual meanings.

■ Question 1 of 1 [CO4] [20 marks + 4 marks (bonus)]

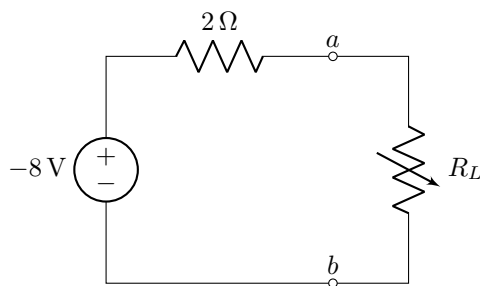


- (a) [16 marks] Find Thevenin's equivalent voltage and resistance for the above circuit with respect to the terminals a and b . (with appropriate $\pm$ sign and unit)

$$V_{Th} = -8 \text{ V}$$

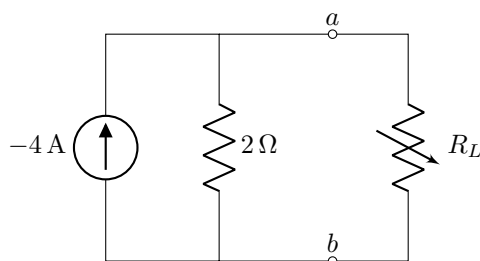
$$R_{Th} = 2 \Omega$$

- (b) [2 marks] Draw the Thevenin's equivalent circuit. Also, identify the terminals a and b in your drawing.



Solution:

- (c) [2 marks (bonus)] Draw the Norton's equivalent circuit. Also, identify the terminals a and b in your drawing.



Solution:

- (d) [2 marks] What should be the value of R_L such that the maximum amount of power can be transferred to the load? (with appropriate $\pm$ sign and unit)

$$R_L = 2\ \Omega$$

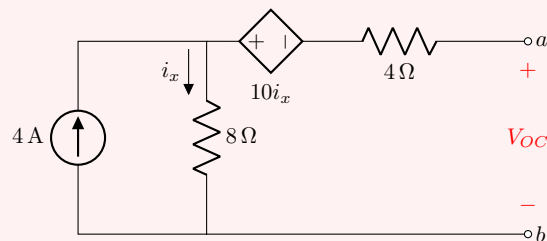
- (e) [2 marks (bonus)] What the maximum amount of power that can be transferred to the load? (with appropriate $\pm$ sign and unit)

$$P_{L_{max}} = 8\ \text{W}$$

Solution (a):

Step 1: Find V_{OC}

In order to find V_{OC} (open circuit voltage), we first need to open the load and then measure the voltage across those terminals.

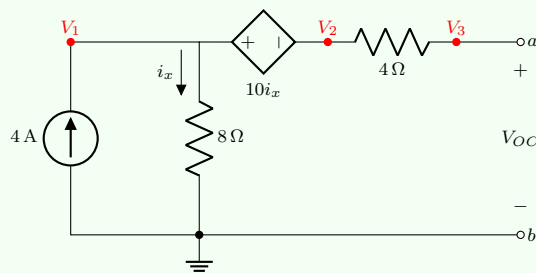


Let's use **Nodal Analysis** to figure out the voltage. The steps of nodal analysis are:

Step 1: Node Identification

There are **3 nodes** besides the ground node. Here, $V_{OC} = V_3 - 0$.

P.S. Since there is no current through the $4\ \Omega$ resistor (in series with the open circuit), there should be no voltage difference across it. So, $V_2 - V_3 = 4\ \Omega \times 0\ \text{A} \Rightarrow V_2 = V_3$. Even if you don't realize this fact (this solution assumes you don't), you should get the correct result anyway.



Step 2: Voltage Source Equation

Here we have a single voltage source. So, we get,

$$\begin{aligned} V_1 - V_2 &= 10 i_x \\ \Rightarrow V_1 - V_2 &= 10 \frac{(V_1 - 0)}{8} & [\because i_x = \frac{V_1 - 0}{8}] \\ \Rightarrow V_1 - V_2 &= \frac{5 V_1}{4} & (1) \end{aligned}$$

Step 3: KCL at Nodes

- Since Node 1 has a voltage source connected to it, we can't write KCL equation for that.
- Since Node 2 has a voltage source connected to it, we can't write KCL equation for that.
- **Node 3** doesn't have any voltage source connected to it, so we can write KCL equation here.

$$V_3 \left(\frac{1}{4} \right) - \frac{V_2}{4} = 0$$

$$\Rightarrow V_3 = V_2 \quad (2)$$

P.S. This fact was predicted in "Step 1: Node Identification".

Step 4: KCL at Supernodes

We have 3 nodes, but we only have 2 equations. So we need 1 other equation from supernodes. Here, we have a **Floating Voltage Source** between Node 1 and 2. So we can write a KCL equation at the **Supernode (1,2)**.

$$V_1 \left(\frac{1}{8} \right) - \frac{0}{8} - 4 + V_2 \left(\frac{1}{4} \right) - \frac{V_3}{4} = 0 \quad (3)$$

Step 5: Solve Equations

Solving the equations, we get, $V_1 = 32 \text{ V}$, $V_2 = V_3 = -8 \text{ V}$

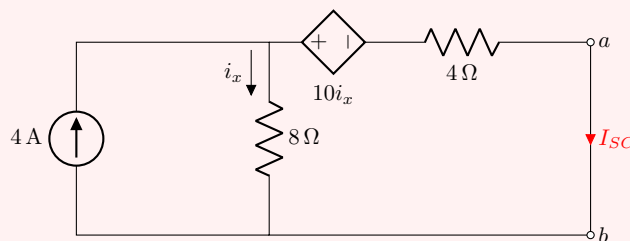
From the nodal analysis circuit, we can see, $V_{OC} = V_3 - 0 = -8 \text{ V}$. So, Thevenin's voltages should be:

$$V_{Th} = V_{OC} = -8 \text{ V}$$

Since V_{OC} is not zero, we need to find I_{SC} (short circuit current).

Step 2: Find I_{SC}

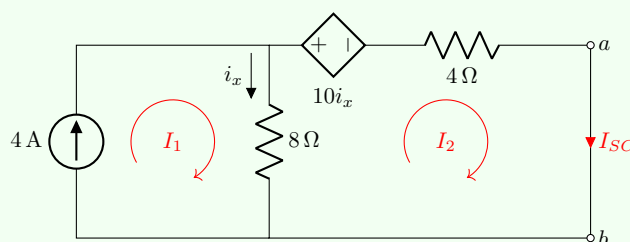
We short the terminals and find the current following passive sign convention with V_{OC} .



Let's use **Mesh Analysis** to find the I_{SC} . Here are the steps of mesh analysis:

Step 1: Mesh Identification

There are **2 meshes** in the circuit. Here, $I_{SC} = I_2$.



Step 2: Current Source Equation

Here we have a single dependent current source. So we get,

$$I_1 = 4 \quad (1)$$

Step 3: KVL at Meshes

- Since Mesh 1 has a current source connected to it, we can't write KVL equation for that.
- **Mesh 2** doesn't have any current source connected to it, so we can write KVL equation there.

$$\begin{aligned} I_2(4 + 8) - 8I_1 + 10I_x &= 0 \\ \Rightarrow 12I_2 - 8I_1 + 10(I_1 - I_2) &= 0 \\ [\because i_x &= I_1 - I_2] \end{aligned} \quad (2)$$

Step 4: KVL at Supermeshes

Not needed, since we have 2 meshes here and we already have 2 equations.

Step 5: Solve Equations

Solving the equations, we get, $I_1 = 4 \text{ A}$, $I_2 = -4 \text{ A}$

From the mesh analysis circuit, we can see, $I_{SC} = I_2 = -4 \text{ A}$. So, we get,

$$I_{SC} = -4 \text{ A}$$

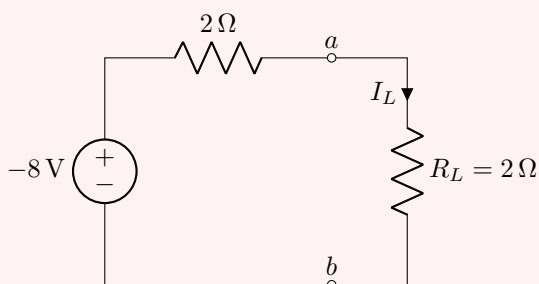
Step 3: Find R_{Th}

We know that, Thevenin's resistance, $R_{Th} = \frac{V_{OC}}{I_{SC}} = \frac{-8}{-4} \Omega = 2 \Omega$

$$R_{Th} = 2 \Omega$$

Solution (d) & (e):**Maximum Power Transfer**

We can transfer the maximum amount of power when, $R_L = R_{Th} = 2 \Omega$. Meaning, when the load resistance equates Thevenin's resistance. We can find the maximum amount of transferrable power by finding how much power the load resistance is consuming in such a scenario.



$$\begin{aligned} P_{L_{max}} &= I_L^2 R_L \\ &= \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \\ &= \left(\frac{V_{Th}}{R_{Th} + R_{Th}} \right)^2 R_{Th} \quad [\because R_L = R_{Th}] \\ &= \frac{V_{Th}^2}{4R_{Th}} = \frac{(-8)^2}{4 \times 2} \text{ W} \\ &= 8 \text{ W} \end{aligned}$$